

AI AML Platform — Interface & API Endpoints

Complete mapping of UI visual blocks, Plotly charts, graph networks, and AI narratives to backend REST services.

TARGET BASE URL	FRONTEND REPO	BACKEND SERVICE	PROTOCOL
http://localhost:8000	AI-AML-FrontEnd	Smart_AML2025 (FastAPI)	REST JSON / Postman

1. Base URL & Dynamic Routing Architecture

The frontend automatically resolves the backend URL via [src/services/apiConfig.ts](#) based on the browser host:

- Server / Local dev: **http://localhost:8000**
- LAN / Intranet: **http://<machine-ip>:8000**
- Environment overrides: **NEXT_PUBLIC_API_URL** or **NEXT_PUBLIC_API_PORT=8000**

2. Investigator Alert Cockpit Screen

Route: `/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}`

UI Visual / Block	Method	API Endpoint & Parameters	Payload / Data Delivered	Component
Customer 360 Glance (Baseball Card)	GET	<code>/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}</code> Path: country_code, inst_code, cus_num, day_date	customer_360_snapshot object (KYC profile, risk level, monthly debit/credit turnover, accounts, high-risk flags)	<code>Customer360Glance.tsx</code>
Forensic AI Translation (Macro Narrative Display)	GET	<code>/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}</code> Path: country_code, inst_code, cus_num, day_date	alert_metadata.ai_generated_narrative (Markdown formatted text)	<code>MacroContextZone (MarkdownRenderer.tsx)</code>
Run AI Forensic Translation (On-Demand Macro LLM Trigger)	POST	<code>/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/generate-narrative</code> Path: country_code, inst_code, cus_num, day_date	Response: {"country_code": "...", "cus_num": "...", "narrative": "### Forensic Summary...", "status": "generated"}	<code>AIThinkingButton.tsx</code>
Transaction Amount Timeline (90-Day Baseline Spline)	GET	<code>/api/history/{country_code}/{inst_code}/{cus_num}/all-transactions</code> Path: country_code, inst_code, cus_num	[{"tra_date": "2025-03-01", "amount": 1250.0}, ...]	<code>PlotlyWrapper.tsx (VisualEvidenceZone)</code>
Timeline Suspect Alerts (Red Pulsing Anomaly Dots)	GET	<code>/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions</code> Path: Params Query: scenario_code=0	[{"TRANSACTION_KEY": "...", "EQU_TRA_AMT": 6111.0, "TRA_DATE": "2025-05-16", ...}]	<code>PlotlyWrapper.tsx (VisualEvidenceZone)</code>
Semantic Cluster Projection	GET	<code>/api/viz/clusters</code> Query: target_cus_num={cus_num}	{"target_node": {...}, "peer_nodes": [...], "global_nodes": [...],	<code>ClusterScatterPlot.tsx</code>

(UMAP 2D Scatter Plot)			"declared_sector": {...}}	
Customer Mini-Profile Drawer (UMAP Peer Node Click)	GET	/api/customers/{cus_num}/mini-profile Path: cus_num (from clicked UMAP node)	{"customer_profile": {...}, "xai_context": {"primary_anomaly_drivers": [...], "recent_transactions": [...]}	ClusterScatterPlot.tsx (Slide Drawer)
Network Syndicate Graph (Neo4j Subgraph Viewer)	GET	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/network Path: Params Query: max_hops=3 (range 1-4)	{"nodes": [{"id": "...", "label": "Customer Account", "isRisk": false}], "edges": [{"source": "...", "target": "...", "relationship": "TRANSFERRED_TO"}]}	NetworkGraphViewer.tsx (ForceGraph2D)
Transaction Ledger Grid (Micro Context Table)	GET	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions Path: Params Query: scenario_code=0	List of MicroPayload with EQU_TRA_AMT, TRA_DATE, DEB_CRE_IND, CONFIDENCE_SCORE, AI_MICRO_PAYLOAD	MicroLedgerZone (Table Grid)
Local Anomaly Drivers Panel (Ledger Side Inspector)	GET	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions Path: Params Query: scenario_code=0	Parsed from selected item: AI_MICRO_PAYLOAD.local_anomaly_drivers (e.g., Unusually high volume, Round amount)	MicroLedgerZone (Detail Panel)
Generate AI Micro Insight (Transaction LLM Trigger)	POST	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions/{tra_seq1}/{tra_seq2}/generate-micro-narrative Path: country_code, inst_code, cus_num, day_date, tra_seq1, tra_seq2	Response: {"narrative": "The transaction of \$6,111 on 16-MAY-2025 exhibits...", "status": "generated"}	AIThinkingButton.tsx (Ledger Detail)
Money Flow Sankey (Directional Cashflow)	GET	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions Path: Params Query: scenario_code=0	Aggregated client-side from DEB_CRE_IND (1=Debit, 2=Credit) and COUNTERPARTY_NAME	MoneyFlowSankey (Plotly Sankey)

3. Supervised Alert Triage Queue Screen

Route: /triage/[country_code]/[inst_code]/[day_date]

UI Visual / Block	Method	API Endpoint & Parameters	Payload / Data Delivered	Component
Triage Alert Feed (Left Queue List)	GET	/api/alerts/{country_code}/{inst_code}/{day_date}/triage Path: country_code, inst_code, day_date	List of TriagePayload objects containing CUS_NUM, TRA_SEQ1, TRA_SEQ2, AI_SUPERVISED_PAYLOAD, customer_360_snapshot	AlertRow (page.tsx)
Risk & Confidence Dial (Radial SVG Gauge)	GET	/api/alerts/{country_code}/{inst_code}/{day_date}/triage Path: country_code, inst_code, day_date	ai_verdict.confidence_score (e.g., "94%", rendered as dynamic circular progress)	ConfidenceArc (page.tsx)

Triage Customer 360 Glance (Profile Snapshot)	GET	/api/alerts/{country_code}/{inst_code}/{day_date}/triage Path: country_code, inst_code, day_date	row.customer_360_snapshot (Profile, Turnover, Risk Score, Sanction Matches)	Customer360Glance.tsx
Supervised AI Verdict Card (Rule Conflict Analysis)	GET	/api/alerts/{country_code}/{inst_code}/{day_date}/triage Path: country_code, inst_code, day_date	ai_verdict.decision ("FALSE POSITIVE" "VIOLATION"), legacy_context.rule_triggered, explanation.top_drivers	Verdict Panel (page.tsx)
Generate Supervised Narrative (Audit Trail Report)	POST	/api/alerts/{country_code}/{inst_code}/{cus_num}/{day_date}/transactions/{tra_seq1}/{tra_seq2}/generate-supervised-narrative Path: country_code, inst_code, cus_num, day_date, tra_seq1, tra_seq2	Response: {"narrative": "Detailed regulatory justification for closing alert...", "status": "generated"}	Button (page.tsx)

4. Root Command Center Screen

Route: /

The root landing page ([src/app/page.tsx](#)) functions as the navigation hub:

- **Omni-Search Bar:** Captures Customer Number (e.g. 100376) and routes to ``/alerts/{country}/{inst}/{cus_num}/{day_date}``.
- **Batch Date Picker:** Sets formatted batch date (e.g. 16-MAY-2025) passed to all alert queries.
- **Launchpad Card:** Launches the Supervised Triage Queue with the active institutional context.

5. API Quick Reference Matrix (For Postman & Interface Team)

HTTP Method	Endpoint URI	Backend Service Handler
GET	/api/alerts/{country}/{inst}/{cus_num}/{day_date}	alerts.py :: fetch_alert_summary
GET	/api/alerts/{country}/{inst}/{cus_num}/{day_date}/transactions	alerts.py :: fetch_alert_transactions
GET	/api/alerts/{country}/{inst}/{day_date}/triage	alerts.py :: fetch_triage_queue
GET	/api/history/{country}/{inst}/{cus_num}/all-transactions	alerts.py :: fetch_transaction_history
GET	/api/alerts/{country}/{inst}/{cus_num}/{day_date}/network	alerts.py :: fetch_customer_network (Neo4j)
GET	/api/viz/clusters	visualization.py :: get_cluster_visualization
GET	/api/customers/{cus_num}/mini-profile	investigation.py :: get_customer_mini_profile
POST	/api/alerts/.../generate-narrative	alerts.py :: generate_narrative (LM Studio)